



CareLink

Agentic AI-Powered Medication Adherence & Chronic-Care Coordination System

Preliminary FYP Concept Report

Degree: BS Computer Science

Project Type: Final Year Project

Team Size: 3 Members

Project Domain: Agentic Artificial Intelligence, Healthcare, Machine Learning, Software Engineering, Mobile/Web Systems

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1. Executive Summary

CareLink is an **agentic AI-powered medication adherence and chronic-care coordination system** designed primarily for elderly patients and individuals managing multiple medications or long-term conditions.

Medication management is not only a reminder problem. Patients may have complex schedules, changing prescriptions, repeated missed doses, refill issues, and difficulty understanding instructions. Family members may know that a problem exists but lack a structured way to monitor it, while healthcare professionals cannot realistically review every routine event in a busy care environment.

CareLink focuses on a computing problem beyond reminders: **continuously interpreting medication-related information and behavior, estimating adherence risk, selecting a proportionate intervention, and coordinating support among patients, caregivers, and nurses through specialized software agents.**

The system will combine medication management, prescription/document understanding, adherence tracking, machine-learning-based risk estimation, adaptive interventions, caregiver and nurse task prioritization, medication-history reconciliation, and secure longitudinal records.

The project does not attempt to diagnose diseases, prescribe medication, or replace healthcare professionals. AI outputs remain supportive and policy-constrained, while clinically important decisions remain under human control.

2. Background and Motivation

Medication adherence remains a major challenge in chronic care, particularly for older adults and patients managing multiple medicines. Recent reviews continue to investigate digital adherence interventions and machine-learning approaches for predicting future non-adherence, with a growing emphasis on prospective risk prediction and clinically actionable intervention rather than retrospective reminder logs alone. [1][2][3]

The market already contains mature medication-management products. Medisafe currently advertises personalized reminders, caregiver coordination, and a machine-

learning JITI predictive engine for non-adherence risk and just-in-time interventions. CareClinic provides medication reminders, adherence statistics, health tracking, caregiver sharing, AI insights, appointments and reports. MyTherapy provides reminders, medication plans, refills, health measurements, family profiles and reports. Hero combines medication management with a connected smart dispenser and caregiver notifications. [4][5][6][7]

Therefore, CareLink does not claim novelty for medication reminders, caregiver alerts, or adherence prediction in isolation. Its proposed research direction is the **agentic coordination of heterogeneous medication evidence, adherence risk, intervention selection, and human escalation**, evaluated against a fixed-rule baseline.

3. Problem Statement

Patients, particularly elderly individuals and those managing multiple medications, may experience difficulties such as:

- Forgetting scheduled medications.
- Taking medications at the wrong time.
- Losing or misinterpreting prescription information.
- Managing several medicines simultaneously.
- Forgetting follow-up appointments.
- Failing to communicate medication-taking patterns to healthcare professionals.
- Depending on family members or caregivers to manually monitor medication routines.

Existing medication-reminder applications commonly focus on isolated functions such as alarms and medication lists. Such functionality does not fully address the broader problem of **medication adherence, longitudinal monitoring, caregiver involvement, and communication with healthcare professionals**.

The problem addressed by CareLink is therefore:

How can an agentic, privacy-conscious computing system continuously interpret medication-related data and behavioral signals, estimate adherence risk, select

proportionate interventions, and coordinate support among patients, caregivers, and nurses without replacing clinical decision-making?

The project must balance useful prediction against privacy, safety against notification fatigue, automation against human oversight, and detailed monitoring against elderly usability.

4. Proposed Research Contribution and Novelty

CareLink acknowledges that existing commercial systems already provide many conventional medication-management features, and that at least one market leader now publicly advertises predictive non-adherence risk and just-in-time interventions. [4]

The proposed contribution is therefore not a claim of inventing medication reminders or prediction. The central research contribution is an **agentic care-coordination loop**:

Observe → Interpret → Predict → Select Intervention → Escalate When Necessary → Observe Outcome → Adapt

The project will investigate whether specialized software agents can coordinate this loop more effectively and transparently than a fixed-rule reminder workflow.

Specific contributions to investigate include:

- Specialized agents for medication management, adherence analysis, intervention selection, and care coordination.
- Human-workload-aware caregiver/nurse escalation instead of forwarding every event to a human.
- Explainable intervention decisions showing the behavioral evidence behind significant alerts.
- Medication reconciliation across historical prescriptions, with human verification of discrepancies.
- Explicit safety and policy controls limiting autonomous agent actions.

5. Purpose of the Project

The purpose of CareLink is to develop a secure and intelligent healthcare-support platform that:

1. Simplifies medication management for patients.
2. Reduces the likelihood of missed medication doses.
3. Allows caregivers to support authorized patients remotely.
4. Detects medication adherence patterns.
5. Converts prescription information into structured medication schedules.
6. Provides healthcare professionals with summarized adherence information.
7. Supports follow-up and continuity of care.
8. Demonstrates practical applications of AI in healthcare while maintaining human oversight.

6. Product Vision

Vision Statement

To make medication and chronic-care support proactive rather than reactive by using agentic AI to identify emerging adherence problems, coordinate appropriate support, and reduce unnecessary burden on families and healthcare workers.

CareLink aims to connect three major participants:

Patient



Caregiver



Healthcare Professional

The platform will maintain a longitudinal view of medication schedules, adherence behavior, relevant health information, and follow-up activity.

7. Project Goals

The primary goals of the project are:

Goal 1 — Medication Management

Provide patients with a structured medication-management system capable of handling multiple medicines, doses, schedules, and durations.

Goal 2 — Intelligent Reminders

Provide customizable and adaptive medication reminders rather than relying only on fixed alarms.

Goal 3 — Prescription Digitization

Allow users to upload prescription images/documents and use AI-assisted extraction to convert prescription information into structured records.

Goal 4 — Adherence Monitoring

Record medication-taking behavior and analyze historical adherence patterns.

Goal 5 — Caregiver Support

Allow patients to authorize trusted caregivers who can monitor adherence and receive configurable alerts.

Goal 6 — Nurse and Care Coordinator Support

Provide nurses or designated care coordinators with a prioritized queue of patients and events requiring attention rather than requiring continuous manual monitoring.

Goal 7 — AI-Assisted Insights

Use machine learning and AI selectively to generate useful, explainable insights.

Goal 8 — Accessibility

Design the system for elderly and less technically confident users using simple navigation, readable interfaces, voice interaction where feasible, and multilingual support as a potential enhancement.

8. Non-Goals and Boundaries

To maintain safety and realistic scope, CareLink will **not**:

- Replace a doctor or pharmacist.
- Independently diagnose diseases.
- Prescribe medicines.
- Recommend that a patient start, stop, or change a medication without healthcare-professional authorization.
- Automatically alter a prescription.
- Claim medical certainty from AI predictions.
- Provide emergency medical treatment.
- Operate as an official hospital or government medical-record system.

AI-generated outputs will be presented as **supportive information or risk indicators**, not definitive medical decisions.

Any clinically significant alert should remain subject to appropriate human review.

9. Target Users and System Roles

9.1 Patient

The primary user.

Capabilities may include:

- Create/manage profile.
- Add medications.
- View medication schedule.
- Receive reminders.
- Confirm medication taken/skipped.
- Upload prescription.
- View adherence history.
- Manage appointments.
- Communicate authorized information with caregivers/health professionals.

9.2 Caregiver

An authorized family member or caregiver.

Capabilities:

- View medication adherence of linked patient.
- Receive missed-dose alerts according to patient-defined settings.
- View upcoming appointments.
- Receive important care-management notifications.
- Assist with medication records.

Patient authorization must be required before caregiver access is granted.

9.3 Nurse / Care Coordinator

The primary professional monitoring role in the proposed workflow.

Capabilities:

- View authorized patient information.
- Review medication history.
- View adherence statistics.
- Review AI-generated observations.

- Compare previous and current medication schedules.
- Review prioritized adherence cases.
- Review AI-generated observations and supporting evidence.
- Record follow-up actions.
- Confirm or correct relevant medication records.
- Escalate selected cases to a physician when required.

9.4 Doctor / Prescriber

An optional downstream professional role rather than a continuously active dashboard user. CareLink may provide concise summaries or escalation information when a case requires clinical attention.

9.5 System Administrator

Responsible for:

- User management.
- Role management.
- Security configuration.
- Audit monitoring.
- System maintenance.
- Technical administration.

10. Proposed Solution

CareLink will provide a unified digital platform consisting of:

Patient Application / Portal

For medication management and daily interaction.

Caregiver Portal

For authorized monitoring and support.

Nurse / Care Coordinator Dashboard

For reviewing prioritized adherence risks, intervention history, and follow-up tasks.

Agentic AI Layer

For medication information interpretation, adherence analysis, intervention selection, and care coordination under explicit policies.

Secure Backend

For authentication, authorization, data management, scheduling, notifications, and audit logging.

11. Core Functionalities

10.1 Medication Management

Users can create medication records containing:

- Medicine name.
- Dosage.
- Frequency.
- Administration time.
- Duration.
- Instructions.
- Prescribing physician.
- Start date.
- End date.
- Current status.

10.2 Prescription Upload and Extraction

The user can upload a prescription image or document.

The system performs:

Prescription Image

↓

OCR / Document Processing

↓

Information Extraction

↓

Structured Medication Data

↓

User Confirmation

The user must verify extracted information before it becomes an active medication schedule.

This human confirmation step is necessary because incorrect extraction of dosage or medication information could create safety risks.

10.3 Medication Reminder System

The system generates reminders according to the medication schedule.

Example:

8:00 PM — Metformin 500 mg

Scheduled after dinner.

Available actions:

- Taken
- Skipped
- Snooze
- Unable to take

10.4 Adherence Tracking

The system records medication events and calculates indicators such as:

- Daily adherence.
- Weekly adherence.
- Monthly adherence.
- Missed-dose frequency.
- Most frequently missed medication.
- Most frequently missed time period.

10.5 Adaptive Reminder System

Instead of using identical reminders for every patient, the system may analyze historical behavior.

Example:

A patient consistently misses evening doses.

The system may increase reminder prominence or recommend caregiver notification according to configured rules.

10.6 Caregiver Alerts

If a patient repeatedly fails to confirm an important medication event, the system may notify an authorized caregiver.

Example:

Medication confirmation missing

Evening medication has not been confirmed.

Notification thresholds should be configurable to prevent unnecessary alerting.

10.7 Doctor Dashboard

A healthcare professional may see:

Patient: Ahmed Khan

Medication adherence: 84%

Last 7 days: 91%

Last 30 days: 84%

Most missed period: Evening

Current medications: 5

Recent changes: 1

Upcoming follow-up: 12 September

This converts large quantities of raw activity data into a concise overview.

10.8 Medication and Prescription Change Comparison

The system can compare previous and newer medication records.

Example:

Previous

- Medicine A — 500 mg
- Medicine B — once daily

Current

- Medicine A — 750 mg
- Medicine C — once daily

System highlights:

- Dosage changed.
- Medication discontinued.
- New medication added.

Any medication change should be confirmed against the authoritative prescription record.

10.9 Appointment Management

Optional integrated functionality:

- Doctor directory.
- Appointment booking.
- Appointment reminders.
- Upcoming appointments.
- Past appointments.
- Follow-up reminders.

Appointment functionality will support the broader care-continuity goal rather than being the primary purpose of the project.

10.10 Medication Reconciliation

CareLink can compare medication records across prescriptions and dates to highlight dosage changes, newly added medicines, discontinued medicines that still appear active, duplicate records, and other discrepancies requiring human verification.

10.11 Caregiver / Nurse Escalation

The system will support a tiered escalation model:

No Action → Patient Reminder → Adaptive Support → Caregiver Alert → Nurse Task → Professional Escalation

The purpose is to reduce unnecessary notifications while ensuring persistent or concerning patterns reach an appropriate human.

10.12 Voice-Friendly Interaction

The system may support simple voice interactions for elderly users, such as asking when the next medication is due or confirming that a scheduled dose was taken.

10.13 Multilingual Interaction

English and Urdu should be considered for the user-facing support layer, with Roman Urdu as an optional extension depending on model and dataset feasibility.

12. Agentic AI Architecture

CareLink will use an agent-based architecture rather than treating AI as a single general-purpose chatbot.

Medication Agent

Handles medication-related tasks and prepares verified structured medication information for other agents.

Adherence Agent

Analyzes historical medication events and estimates adherence risk.

Intervention Agent

Selects an appropriate intervention from a predefined safe action set according to patient behavior, preferences, previous responses, and escalation policies.

Care Coordinator Agent

Aggregates significant events and creates prioritized caregiver/nurse tasks instead of forwarding every raw event to a human.

Information Agent

Answers non-diagnostic patient questions using approved medication records and constrained knowledge sources.

Safety / Policy Layer

Applies deterministic rules that override uncertain AI outputs for safety-critical operations. Agents cannot independently prescribe, discontinue, or alter medication.

The agentic loop is:

**Observe → Interpret → Predict → Select Intervention → Escalate When Necessary →
Observe Outcome → Adapt**

13. Artificial Intelligence Components

AI should be used where it solves a real problem.

AI Component 1 — Prescription Information Extraction

OCR and document-understanding techniques can extract information from prescription images.

Potential output:

Medicine: Metformin

Dose: 500 mg

Frequency: Twice daily

Instruction: After meals

User confirmation remains mandatory.

AI Component 2 — Medication Adherence Pattern Detection

A machine-learning model can analyze historical adherence events.

Possible features include:

- Missed-dose frequency.
- Time of day.
- Day of week.
- Number of active medications.
- Recent adherence trend.
- Historical reminder response.
- Previous missed-dose patterns.

The model may generate:

Adherence risk: High

This is not a medical diagnosis; it is a behavioral/care-management indicator.

AI Component 3 — Personalized Reminder Strategy

Based on historical behavior, the system can determine which reminder strategy is more likely to receive a response.

Possible strategies include:

- Standard reminder.
- Repeated notification.
- Voice reminder.
- Caregiver notification.
- Escalated reminder.

This module could form one of the project's main experimental/research components.

AI Component 4 — Natural-Language Health Information Assistant

A controlled AI assistant can answer questions such as:

"Which medicines do I have tonight?"

or

"What does this prescription say?"

The system should use verified structured records where possible and avoid generating unsupported clinical instructions.

AI Component 5 — Adherence Risk Prediction

An optional machine-learning component may predict the probability that a patient will miss a scheduled dose based on historical behavior.

Example:

Predicted adherence risk for next evening dose: 72%

The prediction can be used to trigger additional support mechanisms.

This should be evaluated experimentally rather than presented as a medical certainty.

AI Component 6 — Intervention Selection

Rather than stopping at a risk score, the agentic layer selects an allowed support action according to explicit policies.

Example:

Low risk → Standard reminder

Moderate risk → Personalized reminder / confirmation request

Persistent risk → Caregiver notification

High or persistent risk → Nurse task

The selected action and its supporting factors should be auditable.

AI Component 7 — Medication Reconciliation Intelligence

Document understanding and medication-entity matching can compare prescriptions across dates and identify dosage changes, newly added medicines, discontinued medicines that still appear active, duplicate records, and unresolved discrepancies for human verification.

14. System Workflow

Initial Setup

Patient/Caregiver



Create Account



Add Medical Profile



Upload Prescription / Enter Medication



AI Extracts Information



Patient/Caregiver Confirms



Medication Schedule Created

Daily Workflow

Scheduled medication time



Reminder



Patient response



Taken / Skipped / Snoozed



Event stored



Adherence analytics updated

Intelligent Monitoring

Historical data



AI/ML analysis



Adherence pattern detected



Risk/insight generated



Patient/Caregiver notification



Optional healthcare-professional review

Clinical Workflow

Healthcare professional logs in



Selects patient



Reviews medication history



Reviews adherence analytics



Reviews AI-generated observations

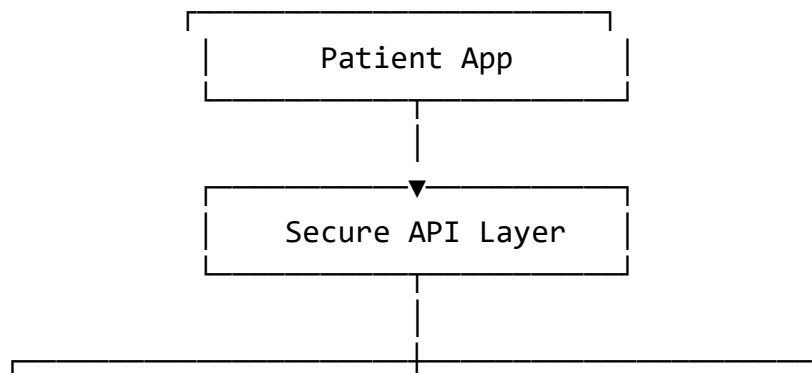


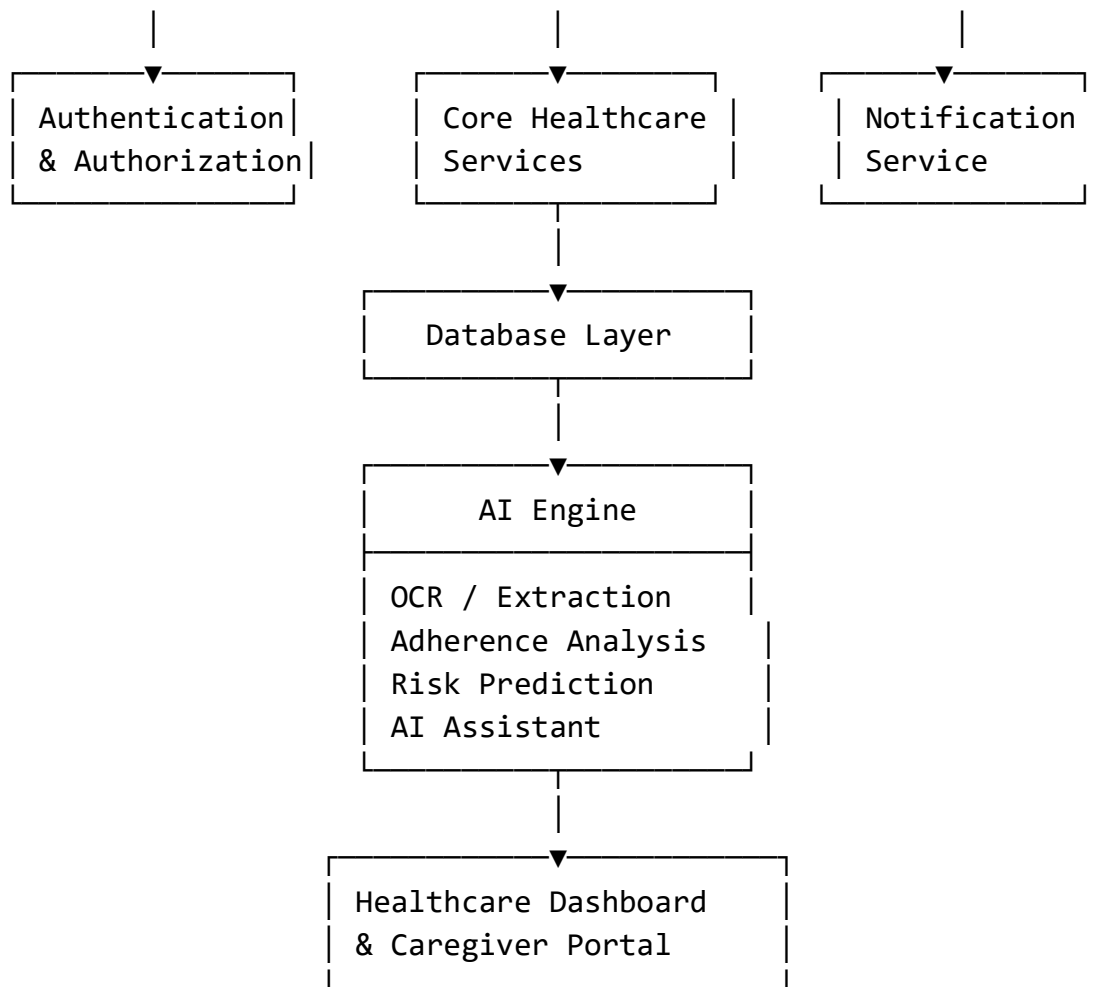
Adds professional assessment / notes



Patient record updated

15. Proposed System Architecture





16. Architectural Principles

16.1 Human-in-the-Loop

AI may analyze and recommend, but important healthcare decisions remain under human control.

16.2 Safety by Design

Medication information extracted by AI must be verified before activation.

16.3 Privacy by Design

Only authorized users should access sensitive patient information.

16.4 Separation of Responsibilities

Patient, caregiver, healthcare professional, and administrator roles must have different permissions.

16.5 Explainability

AI-generated risk indicators should provide understandable reasons or contributing factors where technically feasible.

16.6 Modular Architecture

AI, notification, authentication, database, and healthcare modules should remain independently maintainable.

16.7 Scalable Design

The architecture should support expansion from a student prototype to larger healthcare workflows without redesigning the entire system.

17. Market Comparison and Differentiation

CareLink must be positioned honestly against existing products. The surrounding medication-management features are established; the research contribution is the agentic coordination and evaluation layer.

Capability	Medisafe	MyTherapy	CareClinic	Hero	CareLink
Medication reminders	Yes	Yes	Yes	Yes	Yes, adaptive/policy-driven

Adherence tracking	Yes	Yes	Yes	Yes	Yes, with experimental prediction/intervention evaluation
Caregiver/family support	Yes	Family profiles	Yes	Yes	Yes, with workload-aware escalation
Predictive adherence / JIT intervention	Publicly advertised predictive engine	Not identified as a core public feature in reviewed material	AI insights/pattern detection	Adherence insights	Agentic prediction + intervention coordination
Health/symptom tracking	Yes	Yes	Yes	More limited	Relevant chronic-care data
Prescription / medication records	Yes	Yes	Yes	Yes	Yes, plus reconciliation and verification
Nurse / care-coordinator workflow	Provider/enterprise capabilities	Patient/family-focused in reviewed material	Care-team sharing	Remote-care program	Prioritized nurse task queue
Multi-agent care coordination	Not a publicly stated core architecture in reviewed product material	Not a publicly stated core architecture in reviewed product material	Not a publicly stated core architecture in reviewed product material	Not a publicly stated core architecture in reviewed product material	Central research focus

Explainable intervention decision	Not a stated core research feature	Not a stated core research feature	Insights/correl ations available	Alerts/insi ghts	Explicitly evaluated and auditable
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The market comparison shows why CareLink should not be presented as “another medication reminder app.” Medisafe already advertises predictive non-adherence risk and just-in-time interventions, while CareClinic, MyTherapy and Hero already cover substantial medication, adherence and caregiver functionality. [4][5][6][7]

The differentiating research question is whether **specialized agents can coordinate evidence, intervention selection, caregiver/nurse escalation, and human feedback under explicit safety policies**, with measurable benefit compared with fixed-rule workflows.

18. Functional Requirements

FR-01 — User Registration

The system shall allow users to register accounts.

FR-02 — Authentication

The system shall authenticate users securely.

FR-03 — Role Management

The system shall assign appropriate permissions based on user roles.

FR-04 — Patient Profile

The system shall allow patients to manage their relevant health-management information.

FR-05 — Medication Management

The system shall allow authorized users to add, update, and deactivate medications.

FR-06 — Prescription Upload

The system shall allow prescription documents/images to be uploaded.

FR-07 — Prescription Extraction

The system shall extract relevant prescription information using OCR/AI.

FR-08 — Extraction Confirmation

The system shall require user confirmation before extracted medication data is activated.

FR-09 — Reminder Scheduling

The system shall generate medication reminders according to active schedules.

FR-10 — Medication Confirmation

The system shall record whether a medication was taken, skipped, snoozed, or unconfirmed.

FR-11 — Adherence Analytics

The system shall calculate and visualize adherence statistics.

FR-12 — Caregiver Linking

The system shall allow patients to authorize caregivers.

FR-13 — Caregiver Alerts

The system shall notify authorized caregivers according to configured rules.

FR-14 — AI Analysis

The system shall analyze adherence patterns.

FR-15 — Risk Prediction

The system may calculate adherence-risk indicators based on historical data.

FR-16 — Healthcare Dashboard

The system shall provide authorized healthcare professionals with patient adherence summaries.

FR-17 — Medication History

The system shall maintain historical medication records.

FR-18 — Appointment Management

The system may provide appointment booking and follow-up reminders.

FR-19 — Audit Logging

The system shall record important security-sensitive actions.

FR-20 — Data Export

The system may allow authorized users to export relevant reports.

FR-21 — Agent Task Coordination

The system shall create, assign, track, and close structured agent tasks according to defined policies.

FR-22 — Intervention Selection

The system shall select an allowed support action based on adherence risk, context, and escalation policy.

FR-23 — Medication Reconciliation

The system shall compare relevant medication records across prescriptions and identify discrepancies for verification.

FR-24 — Explainable Alert

The system shall present supporting factors for significant AI-generated risk or escalation decisions.

19. Non-Functional Requirements

Security

- Secure authentication.
- Role-based access control.
- Encrypted communication.
- Secure password storage.
- Access logging.
- Session management.

Privacy

- Minimum necessary data collection.
- Explicit caregiver authorization.
- Controlled healthcare-professional access.
- De-identification of research datasets where applicable.

Reliability

Medication schedules and reminders should remain dependable even during temporary network interruptions.

Performance

Normal application operations should provide responsive interaction and notification scheduling.

Usability

The system should be understandable by elderly users with limited technical experience.

Accessibility

The interface should use:

- Large readable typography.
- Clear controls.
- High-contrast presentation.
- Simple navigation.
- Minimal unnecessary steps.

Maintainability

The software should use modular components, documented APIs, and maintainable code.

Scalability

The architecture should support expansion to more users, healthcare providers, and additional modules.

Explainability

AI outputs should provide interpretable information rather than unexplained predictions wherever feasible.

20. Security, Privacy and Safety

Because CareLink handles potentially sensitive health information, security and privacy will be treated as core system requirements rather than optional features.

Major security measures

- Role-based access control.
- Patient-controlled caregiver authorization.
- Secure authentication.
- Encrypted data transmission.
- Secure database storage.
- Audit logs.
- Token/session management.
- Input validation.
- Protection against unauthorized API access.

AI Safety Measures

The system will:

- Clearly distinguish AI-generated information from professional medical decisions.
- Require confirmation of AI-extracted prescriptions.
- Avoid autonomous prescription changes.
- Avoid unsupported medical claims.
- Prefer structured/verified medication information for safety-critical functions.
- Display appropriate warnings where AI confidence is low.

21. Data Management

Potential data entities include:

Users

- User ID
- Name
- Contact information
- Role

Patient Profiles

- Patient ID
- Relevant health information
- Emergency contact
- Consent settings

Medications

- Medication ID
- Name
- Dose
- Frequency
- Schedule
- Start date
- End date
- Prescribing professional

Medication Events

- Medication ID
- Scheduled time
- Response
- Timestamp
- Status

Prescriptions

- Prescription ID
- Uploaded document

- Extracted information
- Verification status

Caregiver Relationships

- Patient ID
- Caregiver ID
- Authorization status

Appointments

- Patient
- Healthcare professional
- Date/time
- Status

AI Insights

- Patient ID
- Insight type
- Confidence/risk value
- Generated time
- Explanation

22. Scope of the FYP

Core Scope

- Patient medication-management interface.
- Medication scheduling and adherence event tracking.
- AI-assisted prescription/document extraction with human verification.
- Adherence-risk analysis.
- Agentic coordination for medication and adherence tasks.
- Adaptive reminder and escalation policy.

- Caregiver support.
- Nurse/care-coordinator task dashboard.
- Medication-history reconciliation.
- Explainable AI insights.
- Secure authentication, authorization, and audit logging.
- Longitudinal adherence analytics.

Extended Scope

- Voice interface for elderly users.
- Urdu/Roman Urdu support.
- Appointment management.
- Advanced adherence prediction.
- Verified medication-interaction information.
- Wearable integration.
- Additional nurse workflow automation.

These extended features will be implemented only if the core agentic workflow and evaluation are stable.

Scope Boundary

The FYP will not depend on direct integration with national healthcare infrastructure, hospital EMRs, commercial pharmacy systems, or proprietary medical-device APIs unless authorized access becomes available. Evaluation data will use public, synthetic, de-identified, or otherwise ethically approved sources.

23. Complex Computing Problem (CCP)

23.1 Complexity Level Definitions

A **Complex Computing Problem** requires substantial analysis and deeper computing and/or domain knowledge, involves significant interacting constraints or stakeholders,

contains multiple interdependent sub-problems, and does not have one obvious standardized solution.

A **Broadly-defined Computing Problem** can be addressed using established principles and analysis techniques, but involves a wider range of factors, constraints, or context than a routine implementation.

A **Well-defined Computing Problem** can generally be solved using established standards, documented procedures, or standardized techniques with limited ambiguity and relatively few interacting constraints.

CareLink targets the **Complex Computing Problem** category for its complete system, while recognizing that individual sub-functions such as basic reminder scheduling are well-defined components.

23.2 CareLink Mapping to the Nine CCP Characteristics

1. Range of Conflicting Requirements — Complex

CareLink balances privacy against useful prediction, safety against notification fatigue, automation against human oversight, and detailed monitoring against elderly usability.

2. Depth of Analysis Required — Complex

The team must investigate which behavioral signals predict non-adherence, how intervention thresholds should be selected, and whether an agentic workflow provides measurable value over fixed reminder rules.

3. Depth of Knowledge Required — Complex

The solution combines machine learning, agentic AI, databases, APIs, software architecture, privacy/security, human-computer interaction, and medication-management domain knowledge.

4. Familiarity of Issues — Broadly-defined

Medication reminders and adherence are familiar problems. The less routine part is the specific combination of predictive adherence modeling, agentic coordination, explainable intervention selection, and workload-aware human escalation.

5. Level of Problem — Complex

The complete system extends beyond standard CRUD and notification functionality by coordinating AI inference, autonomous task execution, policy enforcement, longitudinal behavior analysis, and human escalation.

6. Stakeholder Involvement — Complex

Patients, caregivers, nurses/care coordinators, doctors as downstream professionals, and administrators have different requirements that can conflict, particularly around privacy, workload, and information visibility.

7. Consequences — Complex

Incorrect prescription extraction, poor risk assessment, inappropriate escalation, or privacy failures can have significant consequences. This requires explicit safety controls and human oversight.

8. Interdependence — Complex

The system forms an interconnected chain:

Medication records → adherence events → AI analysis → agent decision → intervention → response → new evidence → updated analysis.

9. Requirement Identification — Broadly-defined to Complex

The general problem is known, but detailed escalation and intervention requirements require domain investigation and expert input, particularly around when a caregiver or nurse should be involved.

23.3 CCP Statement for Committee Presentation

CareLink addresses a complex computing problem: designing a secure agentic healthcare-support system that interprets medication and behavioral data, estimates adherence risk, coordinates proportionate interventions, and prioritizes human escalation while balancing privacy, safety, usability, AI uncertainty, and caregiver/nurse workload.

24. Feasibility and Implementation Strategy

The project is feasible for a team of three BSCS students because it can be developed as independent modules.

Member 1 — AI / Data

- OCR and prescription extraction.
- Adherence analysis.
- Prediction model.
- AI evaluation.

Member 2 — Backend / Database

- API development.
- Authentication.
- Database.
- Notification services.
- Security.

Member 3 — Frontend / UX

- Patient interface.
- Caregiver interface.
- Nurse/caregiver dashboard.
- Accessibility and usability.

All members will collaborate on integration, testing, documentation, and deployment.

A software-only implementation is sufficient for the FYP prototype; specialized medical hardware is not required.

25. Testing and Evaluation

The project will compare a conventional fixed-rule reminder workflow with the proposed agentic workflow.

Baseline

Fixed medication reminders and manually configured escalation.

CareLink

Risk estimation + agentic intervention selection + policy-based escalation + human feedback.

Metrics

- Adherence-risk prediction performance.
- Precision, recall, F1, and calibration where applicable.
- False-alert rate.
- Missed-dose rate in simulated/evaluated scenarios.
- Number of unnecessary caregiver/nurse escalations.
- Intervention response rate.
- Agent execution latency.
- Number of agent actions per case.
- Human override rate.
- Audit completeness.

The purpose is to determine whether the proposed coordination workflow adds measurable value beyond a fixed reminder system.

26. Expected Outcomes

At completion, the project is expected to demonstrate:

1. A functional medication-management platform.

2. Automated medication reminders.
3. AI-assisted prescription information extraction.
4. Medication adherence tracking.
5. AI-assisted adherence pattern analysis.
6. Caregiver monitoring functionality.
7. Nurse/care-coordinator dashboard.
8. Secure role-based access.
9. Explainable AI-generated adherence and intervention insights.
10. Medication reconciliation support.
11. Agentic coordination with auditable task history.
12. Demonstration using an appropriately de-identified or synthetic dataset.

The system should demonstrate that AI can support medication-management workflows without replacing healthcare professionals.

27. Future Enhancements

Potential future expansion includes:

- Wearable-device integration.
- Smart pill boxes.
- Integration with hospital information systems.
- Pharmacy integration.
- Additional languages.
- Voice-first elderly interface.
- Remote patient monitoring.
- Telemedicine integration.
- Personalized behavioral interventions.
- Integration with national health information systems where authorized APIs become available.
- Advanced longitudinal health analytics.

28. Limitations and Risks

Data Availability

High-quality, appropriately labeled healthcare data may be difficult to obtain. Research experiments should therefore use public/de-identified datasets or synthetic data where appropriate.

AI Accuracy

AI systems can produce incorrect outputs. The project therefore treats AI as a support mechanism rather than an autonomous clinical authority.

Prescription OCR Errors

Poor image quality, handwriting, abbreviations, or unusual formatting may reduce extraction accuracy.

User Compliance

An application cannot guarantee that a patient physically takes a medication merely because the patient marks a dose as taken.

Alert Fatigue

Excessive notifications can cause users to ignore future alerts. Reminder escalation must therefore be carefully designed.

Privacy

Medical information is sensitive, requiring strict access-control and data-protection measures.

Clinical Validation

A university prototype should not be presented as clinically validated software. Clinical validation would require appropriate clinical partnerships, ethics processes, larger datasets, and formal evaluation.

29. Conclusion

CareLink is proposed as an **agentic medication-adherence and chronic-care coordination system**, not merely a medication-reminder application.

The project acknowledges that commercial systems already provide many conventional medication-management functions, and therefore positions its research contribution in the **coordination layer**: specialized agents interpret evidence, estimate risk, select proportionate support actions, create prioritized caregiver/nurse tasks, and learn from outcomes while operating under explicit safety and privacy policies.

The defining principle is:

AI should coordinate support, not replace clinical judgment.

This framing gives the project a clearer computing problem, stronger CCP justification, and an experimentally testable contribution for a BS Computer Science FYP.

30. Preliminary References

1. Digital Health Interventions to Improve Medication Adherence Among Older Adults: A Systematic Review. 2026. [PubMed]
2. Machine Learning Approaches to Predicting Medication Nonadherence: A Scoping Review. 2025. International Journal of Medical Informatics. [PubMed]
3. Artificial Intelligence Approaches to Predicting Treatment Non-Adherence in Chronic Diseases: A Narrative Review. 2026. Frontiers in Digital Health. [PubMed]
4. Medisafe — Patient and Platform Product Documentation, including personalized reminders, caregiver tools, and the JITI predictive engine. [Official Product Documentation]
5. MyTherapy — Medication and Health Tracking Product Documentation. [Official Product Documentation]
6. CareClinic — Medication, Health Tracking, Care-Team, and AI Insights Product Documentation. [Official Product Documentation]

7. Hero — Smart Medication Management and Caregiver Platform Documentation.
[Official Product Documentation]
8. World Health Organization — Medication Safety and Polypharmacy resources.
9. Relevant peer-reviewed literature on agentic AI, healthcare privacy, human-in-the-loop AI, medication reconciliation, and explainable AI to be expanded during the literature-review phase.